State licensing pipeline — handoff documentation

Generated 2026-07-08 from repo commit `8557ac9e` by `python -m pipeline.client.handoff_doc`. Every number below was computed from the repository's actual contents when this was generated — none are hand-maintained. If this document is more than a few weeks old, re-run that command to get current numbers.

For engineers: this document is the plain-English overview. The technical entry points are `CLAUDE.md` (repo root), `pipeline/client/README.md`, and `pipeline/config/README.md`. Everything below is also inspectable in code.

1. What this is, and what you already have

This repository turns state licensing law into structured, comparable data: for each occupation and each state, what license (if any) is required, what it takes to get one, and what the license lets you do. The results are published as an interactive website.

The most important thing to know: **the raw law text is already in this repository.** Collecting statute text from the web is by far the most painful part of this work (see section 6), and we have already done it broadly — 3,202 of the 4,700 possible occupation/state combinations (94 occupations × 50 states, i.e. 68% of the grid) have their law text saved here. Day-to-day work — turning that saved text into published data — **never touches the web-scraping machinery.** You will not need to scrape anything unless you want brand-new material or a refresh of old material.

2. What's easy, what's cheap, and what's hard

The pipeline runs in stages, and they cost very different amounts of money and effort. Current state of the grid, as of generation:

	Count	What it means
Published	13 occupations	Site data built and ready to serve
Extracted, not published	49 occupations	Structured data exists for at least some states; publishing is a free command
Ready to process	2,167 occupation/state cells	Law text on disk; needs only the paid AI steps (~$0.20/cell, ~$424.10 for all of them)
New occupation, not set up yet	285 cells (24 occupations)	Law text on disk, but this occupation has never been processed before — see below
Needs a fetch	19 cells	Law located but text never pulled — requires the scraping setup

A fetch is running right now (checked live at generation time) — some of the 19 cells below are actively being pulled as this document was generated, so that count is falling.

The 13 published occupations: architecture, construction_trades, dentistry, dietetics, electrician, hvac, medicine, nursing, optometry, plumbing, podiatry, professional_engineering, public_accountancy.

In effort terms:

- **Free, minutes:** publish an occupation whose data is already extracted; re-run the deterministic cleanup step.
- **Cheap, no code:** run an already-fetched occupation/state through the two paid AI steps (about $0.20 per occupation/state cell). "Already fetched" means the raw statute text is saved in this repository — the AI steps just haven't been paid for yet. The dashboard (section 3) shows exactly which cells these are and quotes the cost before anything is spent.
- **Medium: bring a brand-new occupation online for the first time.** Before we can extract structured data for an occupation, an AI step has to figure out *what fields matter for it* — a nurse's data looks different from a plumber's (different requirements, different terminology) — and save that as a one-time template (called a "schema" in the code) for the occupation. This is a one-time step per occupation, not per state, and it's a small cost. This is the 285-cell row above: the law is already on disk, but nobody has designed that occupation's template yet. Once it exists, every state for that occupation becomes "ready to process" like any other. Some states may also still need a fetch.
- **Hard, human-supervised:** fetching new or updated statute text from the web (section 6). Plan for a person to be around while it runs.

3. The tools you'll actually use

What you need: Python (the repo ships a ready environment — see `pipeline/client/README.md`), an Anthropic API key for the two paid steps, and access to the site's hosting (Vercel) for publishing. Chrome is needed only for fetching.

Three commands cover everything except fetching (run from the repo root, using the project's Python environment):

1. **Status** — `python -m pipeline.client.status` — recomputes, for every occupation and state, what stage it's at, whether it's current or stale, and what it would cost to finish.
2. **Dashboard** — `python -m pipeline.client.dashboard` — builds an offline web page (`pipeline/results/status/dashboard.html`, open it in any browser) where you tick the occupations you want, see the total cost, and get the exact commands to run.
3. **Runner** — `python -m pipeline.client.run ...` — executes your picks: `clean` (free), `trim` and `extract` (the paid steps — each prints its cost estimate and waits for a typed "yes" before spending anything), `publish` (free).

Nothing in these tools fetches from the web or spends money on its own. The plain-English walkthrough, including re-fetching, is `docs/runbooks/client_refetch.md`.

4. What it costs — and what the costs depend on

Two stages call an AI model and cost money; everything else is free:

Stage	Model used today	Est. cost
Trim — keep only the occupation's parts of the law	`claude-haiku-4-5-20251001`	~$0.05
Extract — turn trimmed text into structured data	`claude-sonnet-4-6`	~$0.15
Schema discovery — design an occupation's data template	`claude-sonnet-4-6`	small, one-time

Three things to understand about these numbers:

- **They are estimates tied to today's models and prices.** The model names above were read from the code at generation time. Anthropic's per-token prices change, and bulk runs go through Anthropic's discounted batch API. Treat the per-cell figures as planning estimates, not invoices — the runner shows the estimate and asks before spending.
- **Swapping models is easy.** Each stage's model is a single setting (a `--model` flag / one constant per stage module). Moving to a cheaper, better, or different vendor's model is a small change; the prompts themselves live in a plain-text config file (`pipeline/config/prompts.yaml`), not in code.
- **The costs are per-run, not per-view.** The website serves previously-built data; people browsing the site cost nothing.

5. Everything adjustable lives in config files, not code

The pipeline's decisions — which occupations exist, how to recognize each one in a state's law, per-state quirks, what the AI is asked, rate limits — are all plain-text YAML files in `pipeline/config/`, each documented in `pipeline/config/README.md`:

- `professions.yaml` — the occupation list and the patterns used to find each one.
- `state_structure.yaml` — per-state source and layout overrides (the single source of truth for "where does this state keep this law").
- `prompts.yaml` — the instructions given to the AI at each paid step.
- `limits.yaml` — every numeric knob (fetch delays, caps, thresholds).

A non-programmer can read these; a technical person can change them without touching Python.

6. Data sources & integrations — the complete inventory

Every occupation and state on the site traces back to exactly two external data sources. That's the whole list — there is no third feed, API, or vendor anywhere in the pipeline. Both are read-only, free, and require no account or contract with either source's operator.

Source	Coverage	Method	Code	Update cadence
Justia (law.justia.com)	98.6% of corpora, 49/50 states	Scrape (real Chrome browser)	`pull_text.py`	~yearly (assumed)
PA Legislature (legis.state.pa.us)	1.4% of corpora, 1/50 states	Scrape (plain HTTP)	`pull_text.py`	~yearly (assumed)

(Coverage measured from the saved files' own recorded URLs, not asserted. Both sources route through the same file, `pipeline/ingest/pull_text.py` — Justia via `fetch_via_browser`, PA via `is_pa_legis_url` and its section parser.)

Both are read-only, free, and need no account or contract with either source's operator. What needs a human, per source:

- **Justia** — a bot-detection challenge needs a **person** to open a real Chrome window and clear it once per session; it can reappear mid-run and pause until re-cleared. If Justia changes its page layout or URL scheme, the parser needs updating.
- **PA Legislature** — no Cloudflare wall, so no person is needed to run it. The only risk is the same as Justia's second case: if PA changes its site's URL scheme or page markup, the parser needs updating.

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Fetching is not fully autonomous — plan to run it yourself, and expect it to need attention. In practice: re-fetching or fetching new statute text isn't a cloud job you kick off and forget — it runs on your machine, it can take hours for a large batch, and every so often the challenge reappears and needs a person to click through it in that browser window. We've built it to restart itself automatically after a crash or if the computer goes to sleep, so it's more resilient than a plain script, but it is not "set and forget." Budget for someone to check on it periodically while it runs.

One more related point: **finding the law is often harder than fetching it.** Many occupations' licensing rules aren't a standalone chapter — they're a few sections buried inside a much broader one (one insurance-agent type inside the entire insurance code). The repo includes automated locator tooling for this, and the per-state mappings it produces are saved in config, so that work doesn't have to be redone.

7. How much to trust the data

- **It's AI-extracted, with checks.** Structured records are extracted by an AI model from statute text, then run through automated consistency checks (`pipeline/eval/check_consistency.py`). It can still be wrong; every record links back to its source law so a human can verify.
- **"Done" doesn't mean "done forever."** Each cell's output depends on its inputs (the law text) and its recipe (prompt + schema + model + code). The repo stamps every output and can tell you whether it's *current*, *stale* (an input or the recipe changed since it was built), or *unstamped* (built before tracking existed). Right now: 750 extracted cells — 350 current, 0 stale, 400 unstamped. The dashboard color-codes this.
- **Some blanks are deliberate.** A hand-verified list (`pipeline/config/verified_absences.yaml`) records occupation/state pairs where the state genuinely has no license — shown as "not state-regulated (verified)," distinct from "we haven't processed this yet."
- **Having a chapter isn't the same as having every relevant law.** Some states put discipline, fees, or hearing rules in shared "umbrella" chapters that apply to many occupations; the repo detects and pulls these, but completeness auditing is ongoing.

Regenerate with `python -m pipeline.client.handoff_doc [--pdf]`.